

Dreaming Neural Associative Memory: A Self-Reorganizing Memory Framework via Competitive Hebbian Replay

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Abstract

Biological brains consolidate memories during sleep, strengthening important information while forgetting unnecessary patterns. However, existing associative memory models maintain stored memories statically, lacking this dynamic reorganization capability. In this paper, we propose Dreaming Neural Associative Memory (DNAM), a two-phase framework combining the Hopfield network, Hebbian learning, Kohonen competitive learning, and BAM. The Day Phase stores input patterns via the Hopfield network, while the Night Phase selectively reorganizes memories and forms associative pairs among surviving memories via BAM.

1. Introduction

Human memory involves not only storage and retrieval but also a dynamic reorganization process. During sleep, we unconsciously replay the experiences of the day, strengthening important memories and discarding unnecessary ones. This sleep-based memory consolidation enables efficient memory management in the brain (Shaham et al., 2022). However, existing associative memory models lack this capability, maintaining stored memories statically. This paper proposes DNAM to bridge this gap.

Problem Definition and Limitations

The Hopfield network, one of the most representative associative memory models, excels at recovering stored memory patterns from incomplete or corrupted inputs (Hopfield, 1982). The synaptic weight between neuron i and neuron j is computed as follows.

$$w_{ij} = \begin{cases} \sum_{m=1}^M y_{m,i} y_{m,j} & i \neq j \\ 0 & i = j \end{cases} \quad (1)$$

Once the weights are computed, they remain unchanged. This is the first fundamental limitation: because memories

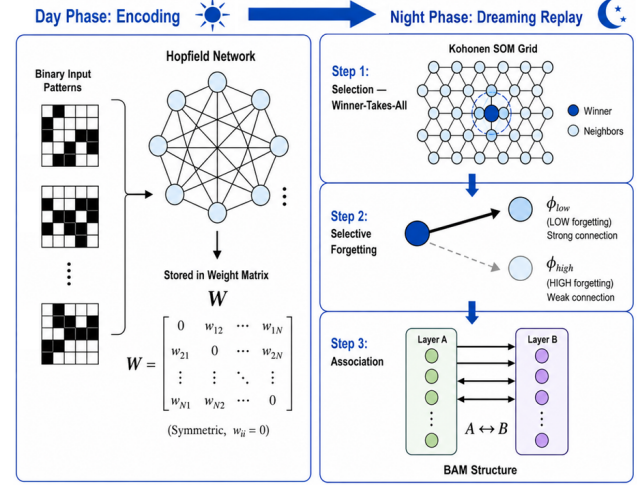


Figure 1. Overview of the DNAM framework: a Day Phase storing patterns via Hopfield network, and a Night Phase reorganizing memories via Kohonen selection, adaptive Hebbian forgetting, and BAM association.

are fixed, the entire weight matrix must be recomputed whenever a new pattern is added. Furthermore, storage capacity is limited, and exceeding it causes the network to converge to incorrect stable states known as spurious states. The Hopfield network also operates only in an auto-associative manner, meaning that given input A, it can only recover A and cannot associate A with a different memory B.

Hebbian learning faces similar problems. According to Hebb's rule, synaptic weights are strengthened when two neurons are simultaneously activated. The weight update rule including the forgetting factor is defined as follows.

$$\Delta w_{ij}(p) = \phi y_j(p) [\lambda x_i(p) - w_{ij}(p)] \quad (2)$$

where ϕ is the forgetting factor and λ is a scaling constant. However, this forgetting factor is a fixed constant applied uniformly to all memories, meaning it cannot distinguish between important and unimportant memories and forgets them all at the same rate. BAM enables hetero-associative recall between pattern set A and pattern set B, but the number of storable associative pairs cannot exceed the number

of neurons in the smaller layer.

Proposed Method: Dreaming Neural Associative Memory (DNAM)

The core philosophy of DNAM is: “**encode by day, re-organize by night.**” It models the principle by which the biological brain replays memories during sleep, retaining only what matters. DNAM operates in a Day Phase and a Night Phase, as illustrated in Figure 1.

Day Phase (Encoding). The Hopfield network stores input patterns as memories. Input data takes the form of binary vectors $Y_m \in \{-1, +1\}^n$, and the weight matrix is computed according to Equation (1). In this phase, all input patterns are stored without discrimination.

Night Phase Step 1 (Selection). The Kohonen network receives the stored memory patterns as input and updates weights based on competitive learning (Kohonen, 1982). The weight correction value is defined as follows.

$$\Delta w_{ij}(p) = \begin{cases} \alpha[x_i - w_{ij}(p)] & j \in \Lambda_j(p) \\ 0 & j \notin \Lambda_j(p) \end{cases} \quad (3)$$

where $\Lambda_j(p)$ denotes the neighborhood of the winner neuron at iteration p . Through repeated learning, patterns that frequently win are classified as important memories, while patterns that never win are classified as unnecessary memories.

Night Phase Step 2 (Selective Forgetting). The key contribution of DNAM lies here. The standard Hebbian forgetting factor is applied uniformly, but DNAM makes it neuron-specific and adaptive.

$$\Delta w_{ij}(p) = \phi(j) y_j(p) [\lambda x_i(p) - w_{ij}(p)] \quad (4)$$

$$\phi(j) = \begin{cases} \phi_{low} & \text{if } j \text{ is a winner or neighbor neuron} \\ \phi_{high} & \text{if } j \text{ is an inactive neuron} \end{cases} \quad (5)$$

Memories frequently selected by Kohonen competitive learning receive ϕ_{low} and their weights are preserved, while memories never selected receive ϕ_{high} and their weights decay rapidly. This replaces uniform forgetting with selective forgetting.

Night Phase Step 3 (Association). The reinforced memory patterns that survive the Night Phase are connected via the BAM weight matrix.

$$\mathbf{W} = \sum_{m=1}^{M'} \mathbf{X}_m \mathbf{Y}_m^T \quad (6)$$

where M' is the number of memories surviving the Night Phase ($M' < M$). This enables hetero-associative recall from memory A to memory B. The overall structure of DNAM is summarized in Table 1.

Table 1. Overview of the DNAM framework.

Phase	Mechanism	Role
Day (Encoding)	Hopfield Network	Store all input patterns
Night 1	Kohonen SOM	Classify important vs. unnecessary
Night 2	Hebbian + $\phi(j)$	Selective forgetting
Night 3	BAM	Hetero-associative pairs

The key contributions of DNAM are threefold: (1) resolving the static weight problem of the Hopfield network via dynamic Night Phase updates, (2) realizing selective forgetting by coupling Kohonen results with the Hebbian forgetting factor, and (3) overcoming auto-associative limitations via BAM to enable hetero-associative recall.

Feasibility and Limitations

The four mechanisms comprising DNAM are each independently well-defined with clear learning algorithms. Therefore, DNAM can be implemented as a sequential pipeline connecting these four components. As a small-scale validation, handwritten patterns such as alphabet characters could be stored as memories, and recovery performance could be tested using noisy or incomplete inputs.

Three limitations are anticipated. First, the phase transition criterion is unclear; the Night Phase can be triggered when the Hopfield network’s energy function E exceeds a defined threshold. Second, ϕ_{low} and ϕ_{high} are subjective; an adaptive forgetting rate that automatically adjusts $\phi(j)$ based on Kohonen neuron activation frequency can mitigate this. Third, binary vector input limits expressiveness for real-valued data; sigmoid activation functions can be introduced to extend the model to continuous-valued inputs.

References

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